

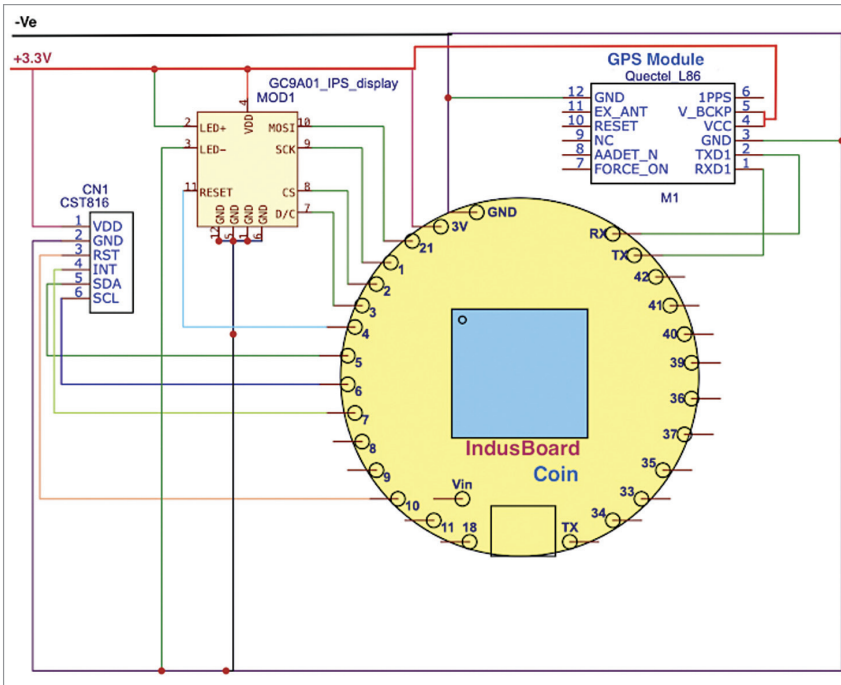


Fig. 3: Circuit diagram of the GPS speedometer for bicycle



Fig. 4: Final GPS speedometer fitted on bicycle handle

drones and robots, which is crucial for autonomous movement and obstacle avoidance. Additionally, it optimises performance in speed bots and motor bots by monitoring speed in various robotic applications.

Fig. 1 shows the author's prototype fitted on a bicycle. For making

this GPS speedometer for the bicycle, the components required are shown in Bill of Materials table.

For coding, first, it is necessary to install the TFT esp8266 and TINY GPS Plus libraries; the next steps are explained in the code. Fig. 2 shows a snippet of the source code.

Circuit and working

Fig. 3 shows the circuit diagram of the GPS speedometer for bicycles and cars. It is built around the IndusBoard, capacitance touchchip CST816, GC9A01A round display, and L86 GPS module.

The L86 GPS uses the serial port with a baud rate of 9600 to send the GPS data. The IndusBoard supports both software and hardware. The L86 GPS pins RXD1 and TXD1 are connected to TX and RX1 of the IndusBoard, respectively. According to the L86 GPS module dataset, the module can be powered from a range of 2.8V to 4V while consuming around 100mA in working mode.

The IndusBoard has four output pins that can handle the current and power. Therefore, the 3.3V and GND pins of the IndusBoard are used for the GPS powering. For more details about the GPS powering system, refer to the previous month's issue of EFY or use its datasheet.

Construction and testing

For the construction of the device, solder all the components according to the circuit diagram (see Fig. 2). After soldering the components together, fix them on top of the bicycle handle or car's dashboard. Now, power the device and start driving.

Based on the change in GPS location with time, it will start giving the speed, which can be seen on the device, as shown in Fig. 4. **EFY**

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